CAMBRIDGE O LEVEL COMPUTER SCIENCE (2210 / 0478)

Paper 2: Algorithms, Programming and Logic | Topic-Wise Question Bank (2024 – 2026 Series)

TOPIC 10: BOOLEAN LOGIC

Exam Sources: Cambridge 2210 / 0478 (May/June 2023, Oct/Nov 2023, May/June 2024)

Question 1 [2210/22/M/J/24 Q2 (2024 Series)] [Total: 6 marks]

(a) State the logic statement for an XOR gate with inputs A and B. **[1 mark]**

(b) Complete the truth table for the logic statement: X = (A AND B) OR (NOT C) **[5 marks]**

A	B	C	A AND B	NOT C	X
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	0	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	1	0	1

✓ Marking Scheme / Model Answer:

(a) XOR statement: Output is 1 if inputs are DIFFERENT (one is 1 and other is 0), otherwise 0 [1 mark].
(b) Truth Table completed accurately: Output column X = [1, 0, 1, 0, 1, 0, 1, 1] [5 marks].

Question 2 [2210/22/M/J/23 Q3 (2023 Series)] [Total: 6 marks]

(a) Complete the 3-input truth table for the logic statement: X = (NOT A AND B) OR (B NOR C) **[5 marks]**

A	B	C	NOT A AND B	B NOR C	X
0	0	0	0	1	1
0	0	1	0	0	0
0	1	0	1	0	1
0	1	1	1	0	1
1	0	0	0	1	1
1	0	1	0	0	0

1	1	0	0	0	0
1	1	1	0	0	0

(b) State which single logic gate has the same output truth table as an AND gate followed by a NOT gate. **[1 mark]**

✓ Marking Scheme / Model Answer:

(a) Truth Table completed accurately: Output column X = [1, 0, 1, 1, 1, 0, 0, 0] [5 marks].
(b) NAND Gate [1 mark].

Question 3 [2210/21/M/J/24 Q3 (2024 Series)] [Total: 5 marks]

(a) Identify the logic gate that produces output 0 only when both inputs are 1. **[1 mark]**

(b) Draw the logic statement representation for the circuit: X = NOT(A OR B) AND (B XOR C).**[4 marks]**

✓ Marking Scheme / Model Answer:

(a) NAND Gate [1 mark].
(b) Logic circuit has three stages:
1. NOR gate with inputs A and B [1.5 marks]
2. XOR gate with inputs B and C [1.5 marks]
3. AND gate connecting outputs of NOR and XOR to give final output X [1 mark].
